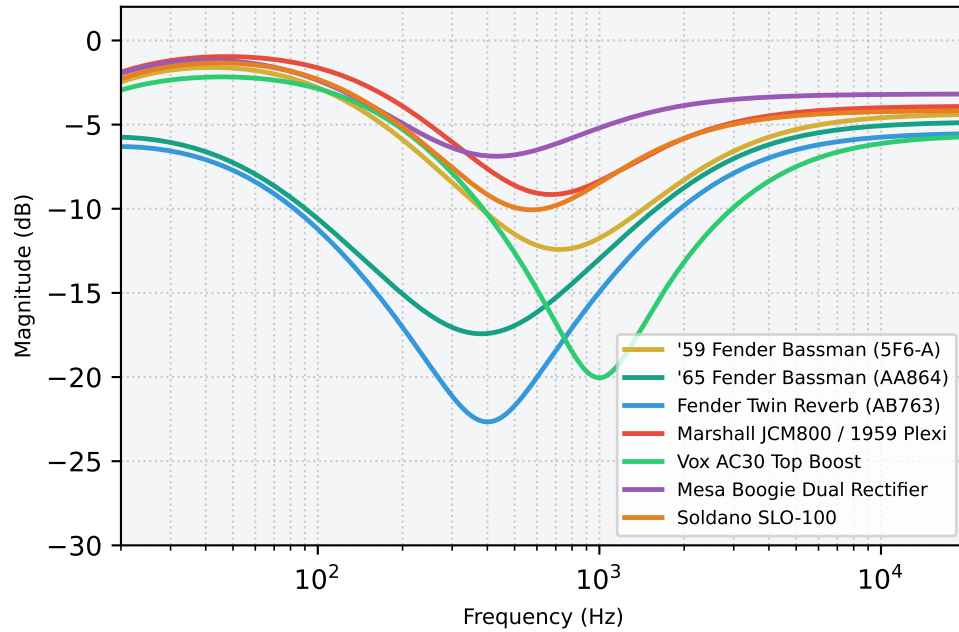


CLASSIC AMP TONE STACK COMPARATIVE BENCHMARK

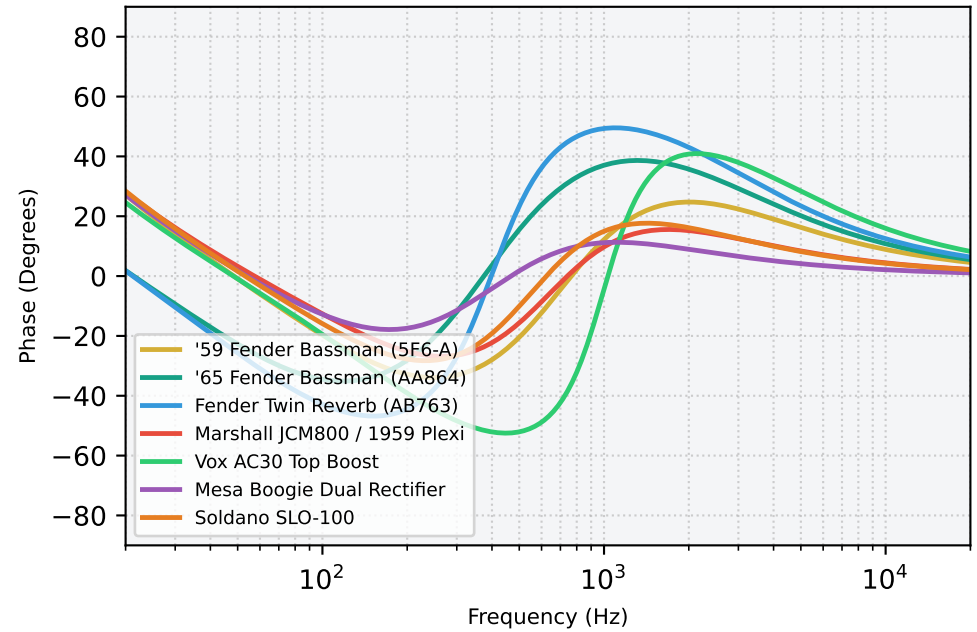
Mathematical Comparison of 6 Iconic Circuit Topologies (Fender, Marshall, Vox, Mesa, Soldano)

Yeh & Smith DAFx-06 Nodal Synthesis

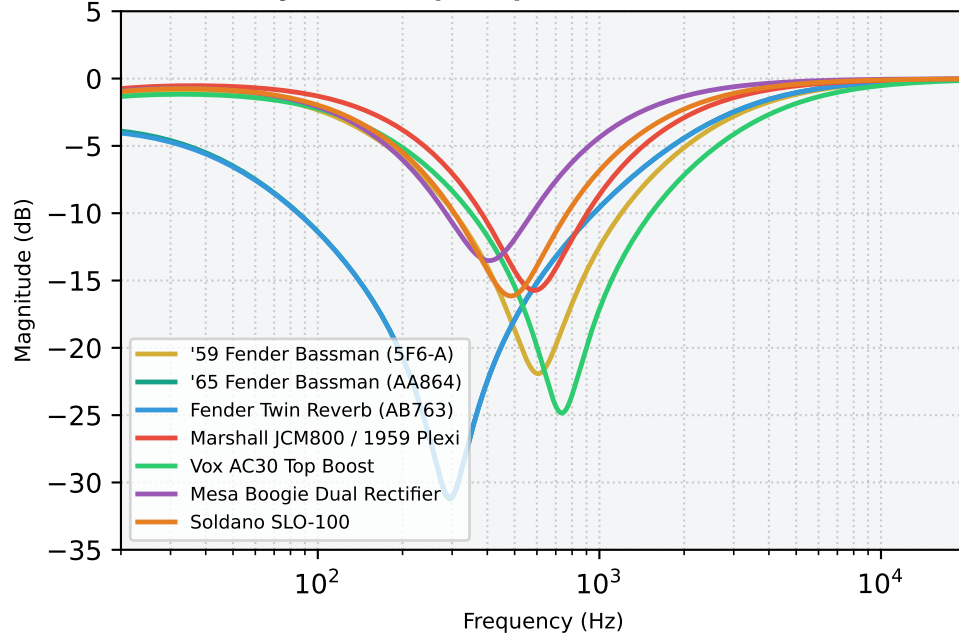
Factory Default Tone Signatures (T=5, M=5, B=5)



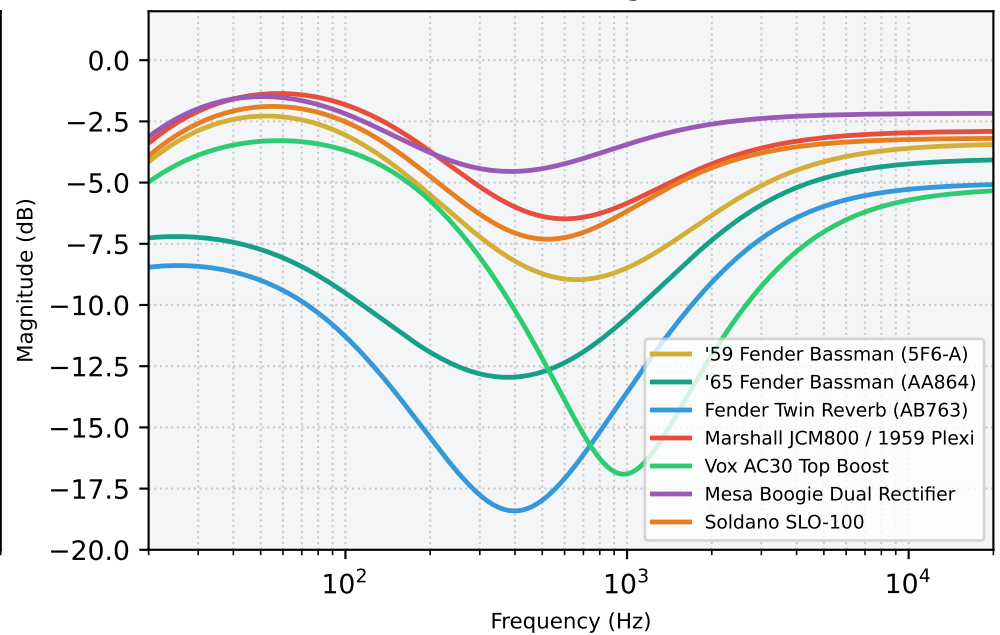
Phase Trajectories across Tone Stacks



Heavy Mid-Scoop Response (T=10, M=0, B=10)



Mid Push / Solo Lead Setting (T=5, M=10, B=3)



CIRCUIT COMPONENT SPECIFICATIONS & ACOUSTIC MAPPING

Mathematical Component Values and Characteristic Notch Frequencies

Ready for C++ Modular Implementation

Amplifier Circuit	R1 (Treb)	R2 (Bass)	R3 (Mid)	R4 (Slope)	C1 (Treb)	C2 (Bass)	C3 (Mid)	Notch Freq	Acoustic Personality
'59 Fender Bassman (5E6)	250k	1.0M	25k	56k	250 pF	20 nF	20 nF	725 Hz	Original Tweed benchmark; balanced warmth, 80% treble, 100% bass
'65 Fender Bassman (A58C)	250k	250k	25k	100k	250 pF	100 nF	47 nF	350 Hz	Blackface Bassman; 100nF deep bass cap, 47nF mid cap
Fender Twin Reverb (A58C)	250k	250k	10k	100k	250 pF	100 nF	47 nF	400 Hz	Classic Blackface; huge low-end authority, deeper scooped mids
Marshall JCM800 / 1959	220k	1.0M	22k	33k	470 pF	22 nF	22 nF	672 Hz	British crunch; forward midrange, higher treble bite
Vox AC30 Top Boost	1000k	1.0M	10k	100k	47 pF	22 nF	10 nF	984 Hz	British chime; 47pF small treble cap, 100k slope, sparkling
Mesa Boogie Dual Rect	250k	1.0M	50k	47k	680 pF	20 nF	20 nF	620 Hz	Modern metal; 680pF treble cap, 50k mid pot for extreme
Soldano SLO-100	250k	1.0M	25k	47k	470 pF	20 nF	20 nF	577 Hz	Boutique gain; legendary singing lead, smooth